

Pranay Saha

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PROFESSIONAL EXPERIENCE

Masterworks , Computer Vision Engineer SDE-1

Mar 2025 – present
Hyderabad, India

- Contributed to the company's flagship low-code platform powering **city-scale surveillance** across highways, parks, and streets, enabling automated incident detection and real-time alerting.
- Architected multiple microservices enabling seamless, one-click deployment of computer vision use cases across distributed edge devices.
- Managed and scaled high-throughput model inference using **Triton Inference Server**, supporting **24x7 concurrent camera streaming** across large deployments.
- Developed a dynamic **multi-accelerator AI inference orchestration service** that converts models into optimised runtime engines for **NVIDIA and Qualcomm hardware**, enabling seamless hardware-specific deployment and inference execution.
- Designed a robust inference architecture capable of processing real-time video feeds from **4,000+ cameras** across heterogeneous edge devices.
- Optimised the end-to-end inference pipeline by eliminating redundant service calls and restructuring data flow, achieving **~70% reduction in computational overhead** and significantly improving overall system throughput.
- Built user-configurable **Computer Vision Use Nodes** (Detection, Annotation, Classification, Alerting), enabling rapid customisation of surveillance workflows without engineering intervention.
- Engineered data pipelines and trained **computer vision detection and classification models** for city surveillance tasks, like entry/exit monitoring and authorised vs. unauthorised access alerting.

Yukin , Machine Learning Engineer

Feb 2024 – Dec 2024
Remote

- Collaborated with the founding team developing models for personalized **Digital Avatars** with **realistic expressions and lip-synchronization**.
- Performed Experiments and Evaluation on **20+** open-source models for **Text2Speech, Speech Conversion, Voice Cloning, and Speech Synthesis** in a multilingual setup, performed optimization to reduce **RAM consumption by 50%**
- Designed and developed a **RESTful API** using **FAST-API** and maintained API documentation and specifications, resulting in a 20% increase in developer productivity.

PROJECTS

Smart Visitor Management System

- Developed a **visitor notification and security system** solution for residential complexes and office spaces to track and monitor all entries.
- Visitors provide basic details at entry, after which the system captures **facial data** and instantly notifies the concerned owner.
- Repeat visitor entries and exits are handled automatically, with all activity clearly **reflected on a live dashboard**.

EDUCATION

B-Tech -CSE(AIML), Institute of Engineering Management
CGPA-8.88

Nov 2020 – May 2024
Kolkata, India

Courses:

Data Structures and Algorithms, Machine Learning, Pattern Recognition, Database Management System, Object-Oriented Programming

SKILLS

Languages — Python, Java • **Tech Stack & Frameworks** — Git, Docker, Kubernetes, PyTorch, TensorFlow, TensorRT, ONNX, Triton Inference Server, LangGraph, Open-CV, Kafka